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School of Computer & Information Sciences GROUP Assignment Coversheet

SECTION 1 This section must be completed by all students

Student ID Number	Surname	Given Names
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NB. For groups larger than 5 please use second coversheet.

Paper Code	COMP728	Paper Name	Internet of Things and Applications
Assignment No. & Title	Literature Review & Survey		
Lecturer Name	Mahsa Mohaghegh		
Due Date	21/04/2023	Submit Date	21/04/2023

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Signatures: Logan Martlew Declan Ross

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The details of the contribution of each member of the group are as follows:

Student 1	
Name: Logan Martlew	Student ID: 19076051
Contributed: Literature and Product reviews, small part of introduction, software architecture, product features	
Signed: Logan Martlew	Date: 21/04/2023
Student 2	
Name: Declan Ross	Student ID: 20108351
Contributed: Literature and Product reviews, introduction, analysis, hardware architecture, conclusion	
Signed: Declan Ross	Date: 21/04/2023
Student 3	
Name: Cassidy Johnstone	Student ID: 17002384
Contributed: Currently nothing due to circumstances, but we will likely resubmit with an SCA application	
Signed:	Date: 21/04/2023
Student 4	
Name:	Student ID:
Contributed:	
Signed:	Date:
Student 5	
Name:	Student ID:
Contributed:	



Auckland University of Technology

Assessment One

Group ID:

The A Team

Students:

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Course name:

Internet of Things and Applications (COMP728)

Submission date:

21/04/2023

1. Introduction

Our smart alarm clock is an alternative to a regular sound-based alarm clock. Our device is able to detect and analyse the owners sleeping habits and determine an optimal sleep and wake-up time. The system is built on the Raspberry Pi platform and utilises multiple input and output components; a speaker for a sound-based alarm, an ultrasonic sensor to detect the user getting out of bed, a PIR sensor for detecting user movement, a microswitch to sense the presence of the user's phone and finally a button for letting the alarm clock know you are in bed.

The basic fundamental features of our prototype include analysing the user's sleep and providing an optimum sleep schedule, forcing the user to get out of bed by not shutting off the alarm clock until they get up and finally, detecting the user's mobile phone by using a microswitch on top of the alarm clock. The device will connect to a web server where the user will access a web app where they can enter their credentials to access their alarm clock. The app will allow the user to access controls and features, statistics and data collected such as their quantity of sleep, quality of sleep and other things. The site will additionally provide a sleep score to the user, gamifying sleep which could incentivise the user to go to sleep earlier.

This product is unique because it separates itself from other generic alarm clocks by using an advanced array of sensors to ensure that the user wakes up with as little sleep inertia as possible, rather than just a sound-based alarm which triggers a fight or flight response. The target market for this product is students and workers as they typically struggle to maintain a healthy sleep schedule.

2. Reviews

2.1. Literature Reviews

The article “The Importance of Sleep” written by the Health Navigator Editorial Team (2022), is a journal article written about the significance of a good sleep schedule, good quality of sleep and the impacts on your health. The article's first point tells us that while we sleep, our bodies recover mentally and physically from the day. The editors also talked about how maintaining a good sleep schedule helps with emotional well-being, physical health, daytime performance and personal safety. The article briefly details sleep stages and how and what the body is doing during different stages of sleep such as; light sleep, deep sleep and REM sleep. We will incorporate this into our design by having the device analyse the users sleeping patterns and waking them up at the most optimum time to maximise their well-being and physical health.

The article “Is a sound-based-alarm clock the best way to wake up?” written by Dr Raminder Mulla (2023) is an article that goes into detail about different ways to wake people up, and how to decrease sleep inertia (How long it takes your body to adapt to waking up). The article focuses on how traditional sound-based alarm clocks are generally effective at getting someone out of bed, but are not a healthy way to wake up. The points of the article are that when someone is asleep, they can be in different states of sleep ranging from light sleep to deep sleep to REM. It states that the deeper the person is in a sleep state, the more sleep inertia they will have when they wake up. They propose using a light-based alarm, or gentle music to wake up the person slower, increasing the probability to wake up the person while they are in a lower sleep stage. Our product will use gentle music as a way to gently wake up the person, by easing into the music. We should be able to only wake the person when they are in their lightest sleep stage.

An article by Chaput et al. (2018) discusses ideal sleeping hours and how they are affected by age. One of the interesting points from the article was that the required sleep hours do not increase with age, but that the ability to get them becomes harder as older individuals tend to have more trouble sleeping and staying asleep. This means that a product like ours may be able to reduce the sleep required for older individuals if it is able to increase their quality of sleep. Another interesting point was that adolescents tend to shift to an evening chronotype and older adults tend to shift to a morning chronotype. This means that adolescents tend to wake up and go to sleep later, while adults tend to wake up and go to sleep earlier. We could use this information to improve our product and assist in the user getting better sleep if we were to collect some information about the user; age, in this case. We could then recommend the user go to sleep earlier if they aren't getting enough sleep and suggest they use their phone less at night if the clock detects that the phone is not present on the charger.

An article by the American Sleep Apnea Association (2023) describes the sleep cycle and its stages, what happens to the body during these stages, as well as describing circadian rhythms and the biological clock. The first 3 stages of sleep are considered NREM stages (Non-REM). The first stage is a short period of around 5 minutes where the person is dozing off and can be easily roused. The next two stages are also NREM stages where the person is becoming more relaxed and brain activity is slowed, and the user eventually enters deep sleep. The final stage is REM sleep, where most dreams occur for people. According to the American Sleep Apnea Association (2023), REM stages tend to make up around 25% of sleep in adults, and they tend to get longer with each sleep cycle. The circadian system in the body produces an alerting signal that indicates whether the person should be awake or not. A few hours before the person's typical bedtime, the circadian system will decrease its alerting signal, and it will increase it a few hours before the person's wake time. This is why we naturally want to sleep and wake up at certain times. We can use this information in our product to determine the ideal time to wake the user, which would be just after REM sleep when they are in the first stage of the sleep cycle and are easily roused. This will make it easier to wake the user, and less jarring.

2.2. Product Reviews

The article “Loftie Alarm Clock Review” is a product review written by Lipski (2023) on an alarm clock sold on Amazon. The product is a clever design, however, it is expensive, difficult to use and does not prevent the user from using their smartphone, only replacing it. The alarm clock has a great many benefits too that our design can utilise such as a gentle tone to wake up the person. The reviewer also claims that Loftie makes falling asleep a lot easier as it generates white noise when it's time to go to sleep. Our design will have a sub-system that will require the user to physically place their phone on top of the alarm clock before going to bed. In addition, our product will have a more straightforward interface than Lofties single-use phone application.

The Raspberry Pi-based Smart Alarm developed by Anwar et al. (2021) uses a microphone and speech analysis to detect sleep cycles, as well as a speaker to wake up the user at an appropriate time. The components were provided as part of the Google Voice Kit. The Smart Alarm has a more advanced system for detecting sleep cycles than some other products as it uses speech recognition which has the potential to be much more accurate and effective. In practice, however, the system was not very good at detecting the sleep cycles of the users as it picked up other sounds and background noise and recognised them as being made by the user. This is a drawback of using advanced and complicated methods of sleep detection. Our prototype uses a much simpler system for calculating sleep cycles as it has a button for the user to tell it when they're going to sleep and calculates their wake-up time based on their average sleep cycle time. This has less potential for optimising the wake-up time, but it is much simpler so should be more consistent and cheaper to produce.

3. Analysis

The purpose of this analysis is to evaluate our prototype alarm clock by analysing pre-existing smart alarm clock designs, as well as research journals and articles relevant to things such as sleep duration, stages, sleep debt and sleep quality. A majority of alarm clocks and phone alarms use sound to abruptly wake people from their sleep. The components of the clocks that we have analysed are sufficient for waking people up but do so in a way that negatively affects the person throughout the day by increasing their sleep inertia. These products are also lacking features that help the user get to sleep and they rarely adapt to a person's sleep schedule or improve it.

We also reviewed several papers about the science of sleep that went into detail about sleep. One of these articles in particular discusses Sleep States and Sleep inertia. They said that while humans sleep, they constantly go through different stages of sleep ranging from light sleep, deep sleep and REM sleep. The higher your sleep state, the more difficult it is to wake from sleep and the more sleep inertia the person will have. The article discusses sound-based alarm clocks and how they cannot know what sleep state you are in, so they will sound a very loud and aggressive alarm to wake you up no matter what. This is effective at waking the person up but has the drawback of causing the person to have a horrible experience in the morning. In addition, we pointed out in our product reviews that they did not force the user to stop using their phone. For example, the Loftie product replaces the need for a mobile phone but it does not restrict the use of the device. Another example of an issue is the Raspberry Pi Smart alarm clock by Anwar et al. (2021) using speech recognition to determine what sleep stage the person was in. In theory, the system was a good idea but unfortunately, in practice, the functionality was limited.

Our smart alarm clock, by comparison, intends to solve these issues. Our product will use gentle music to slowly wake the user up instead of waking them up suddenly and causing a high amount of sleep inertia thus solving the first and third problems. We will solve the second issue by designing our device to require the user to place the phone on top of the alarm clock, this could be done by consistently reminding the user to place their phone on top.

Our IoT setup will involve the device connecting to the internet via Wi-Fi or ethernet and connecting to a web server. The device will utilise an I2C display module, a speaker, an array of sensors, and finally an array of switches and buttons. The chassis of the product will be 3D printed and designed using cad software.

The target market is those who struggle to get good sleep or maintain a good sleep schedule, such as students and full-time workers. Our smart alarm clock aims to provide a solution to those who also have issues with using their phone late at night, by providing incentives such as a 'scoreboard' or 'sleep score' we are able to gamify sleeping.

Cost-effectiveness is also an important factor in our design. The Loftie alarm clock starts at about \$150, which immediately is a major turn-off for users who would want to fix their schedule. Our product will be inexpensive to produce as 3D printing parts is extremely cheap and our parts are mostly plug-and-play.

4. Discussion

4.1. Architecture

The device will sit in a 3D-printed chassis to house all the components such as the raspberry pi, buttons and sensor array. We decided on using a Raspberry Pi because it offers a GUI and a wide variety of input and output devices, (buttons, speakers etc). The sensor array consists of an ultrasonic sensor and a PIR sensor. These sensors work in tandem with each other to detect movements and when the person gets out of bed. The sensor array and buttons will connect to the raspberry pi module using the GPIO (General purpose input output) pins. The design will be made in CAD software and will be simple and modern as this is generally appealing to everybody and fits with most furniture.

Our product will make use of software to support the physical hardware and provide access to the sensors and components to bring together the architecture of our system. For our code that accesses the hardware directly on the Raspberry Pi, we will be using Python. This was an easy decision to make as Python is a very popular choice for developing on Raspberry Pi's and is the most popular language for IoT projects. This will interact with a Node.js web server that will access a database to store data such as user preferences, and wake-up times, as well as data used to provide information to the user such as their sleep score and average sleep cycle time. The web server will also be accessed by a React web application, which will be the user's interface for configuring the alarm clock and viewing the data that is stored and processed on the server. This gives the user a convenient way to configure the alarm as well as view the data gathered by it, and can be accessed in a browser or on their phone. The Python code will live on the Raspberry Pi, while the Node.js web server and React application will be hosted on AWS. Using a cloud provider like AWS to host the application is a good way to bring down costs as you only pay for exactly what resources you use, as well as having the ability to scale up the processing power if more people were to purchase the devices. There will also be no costs involved in software licensing as Python, Node.js and React are all open-source projects and are able to be used commercially for free.

4.2. Product Features

Feature 1 is a web app that the user can access to configure the alarm clock. The primary purpose of the app is to allow the user to set their wake-up time, but there are also some other important functions it provides. The app will allow the user to give a rating of how they felt when waking up, and the app will use this to adjust and calculate the average sleep cycle time of the user. The average time will then be used when calculating wake-up times. The user will also be able to see a history of when they went to sleep and when they woke up. Other feedback will also be given to the user, such as a rating based on their phone usage, and recommendations for what they can do to improve the quality of their sleep.

Feature 2 is a wireless charger that sits on top of the device and has an accompanying phone sensor (microswitch) to detect when the person is not using their phone. It has been well-researched that using screens with blue light is detrimental to sleep quality if used close to a person's bedtime, so our device will encourage the user to put their phone down and discourage the use of it in bed. The app will provide a rating based on the user's phone usage and prompt them to put it down when it is nearing their bedtime.

Feature 3 is motion detection, detecting when the user gets up and out of bed. The sensors on the device will be used to detect when the user gets out of bed in the morning, so that if the alarm goes off and the user dismisses it, the alarm will sound again if the user doesn't get out of bed within a few minutes. This prevents the user from just dismissing the alarm and going back to sleep, which is counterproductive, but it also will stop the user from getting the benefits of the sleep cycle analysis and optimisation, defeating the purpose of the product. This feature will be able to be turned on and off if the user doesn't like it.

Feature 4 is sleep cycle monitoring to optimise the wake-up times to help the user feel less groggy. The alarm will have a prominent button on top of it that will be used to tell the alarm clock when the user is going to sleep. When pressed, the device will calculate what time to wake the user up based on the data collected about their sleep cycles, and based on the time they have set to wake up in the app. The user can also press the button again to re-calculate the wake-up time in case they didn't actually go to sleep when pressing the button the first time. The calculations for waking the user up will take into consideration the data previously collected about the user's sleep cycles, the user's feedback on how they have slept on previous nights, and also based on the literature we have reviewed that gives us a good rough estimate of how long a sleep cycle is.

5. Conclusion

In conclusion, the purpose of this analysis was to analyse articles and product reviews to find more information about good sleeping habits, sleep inertia and sleep stages, and shortcomings of other smart alarm clock designs. It provides us with some insight into the research that has already been done, and what products already exist that we can learn from to improve our own design.

We reviewed journals and product reviews ranging from studies conducted by individuals and organisations to product reviews written by influencers and the general population. By finding out what worked and what didn't in these products and analysing the research papers, we got a more in-depth insight into what we needed to implement into our own smart alarm clock design.

Through our research, we outlined a few issues that we can improve upon with our own design. An example of such is sound-based alarm clocks that are effective at waking up a person, but they do so in a "violent manner" that can cause the person to have bad sleep inertia. We solved this issue in our design by using gentle music to slowly wake up.

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